



DBMS

INTRODUCTION

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What is data?

- ▶ Data are the known facts or figures that have implicit meaning. It can also be defined as it is the
- ▶ representation of facts, concepts or instructions in a formal manner, which is suitable for
- ▶ understanding and processing. Data can be represented in alphabets (A-Z, a-z), digits (0-9) and
- ▶ using special characters (+, -, ., #, \$, etc)
- ▶ e.g: 25, "ajit" etc.

Information:

- ▶ Information is the processed data on which decisions and actions are based.
- ▶ Information can be defined as the organized and classified data to provide meaningful values.
- ▶ Eg: "The age of Ravi is 25"

File & File Oriented Approach

► File:

- File is a collection of related data stored in secondary memory.

File Oriented Approach:

- The traditional file oriented approach to information processing each application has a separate master file and its own set of personal file. In file oriented approach the program dependent on the files and files dependent upon the programs.

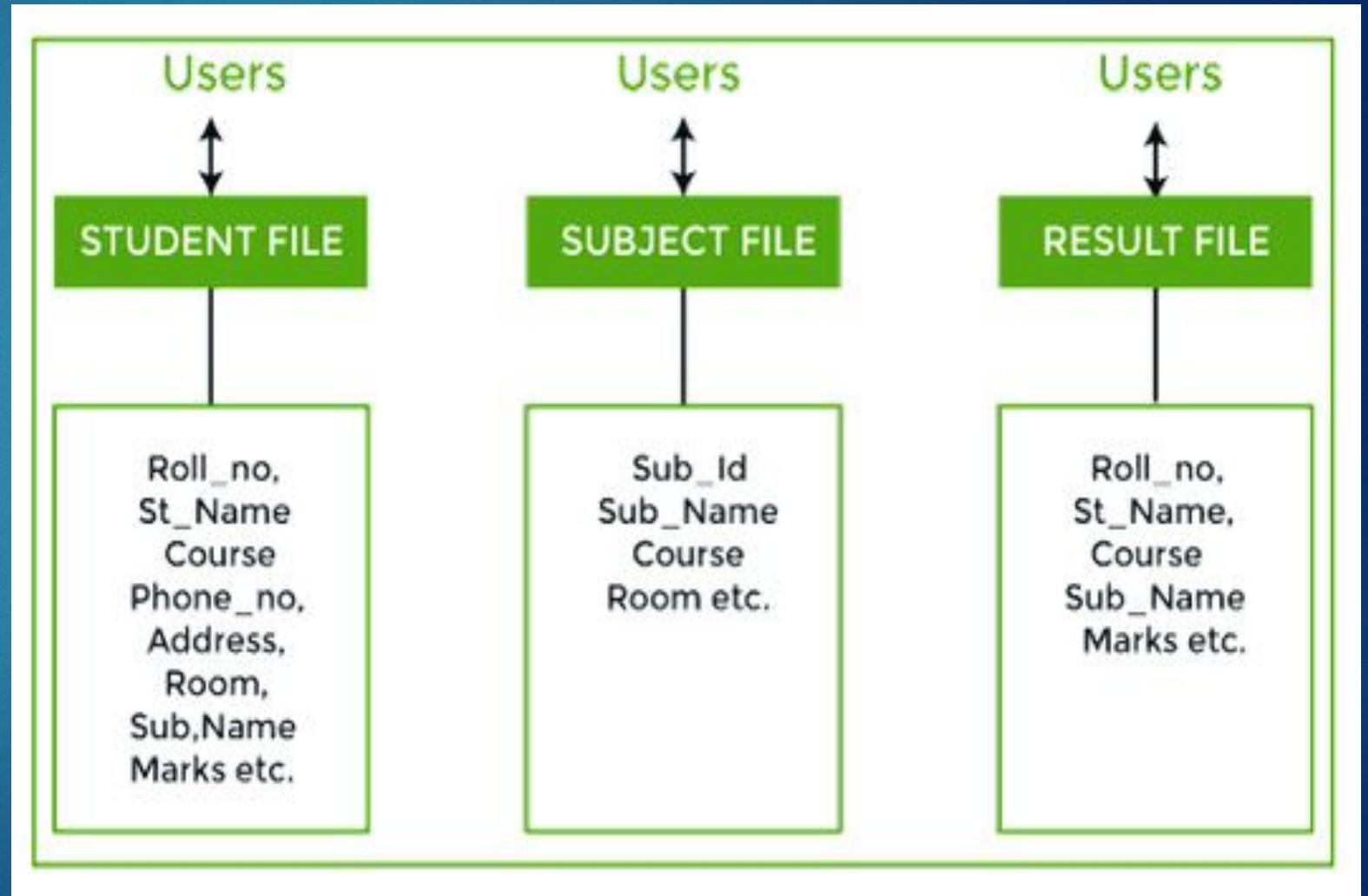
File System Approach

- ▶ File based systems were an early attempt to computerize the manual system. It is also called a traditional based approach in which a decentralized approach was taken where each department stored and controlled its own data with the help of a data processing specialist.
- ▶ The main role of a data processing specialist was to create the necessary computer file structures, and also manage the data within structures and design some application programs that create reports based on file data.

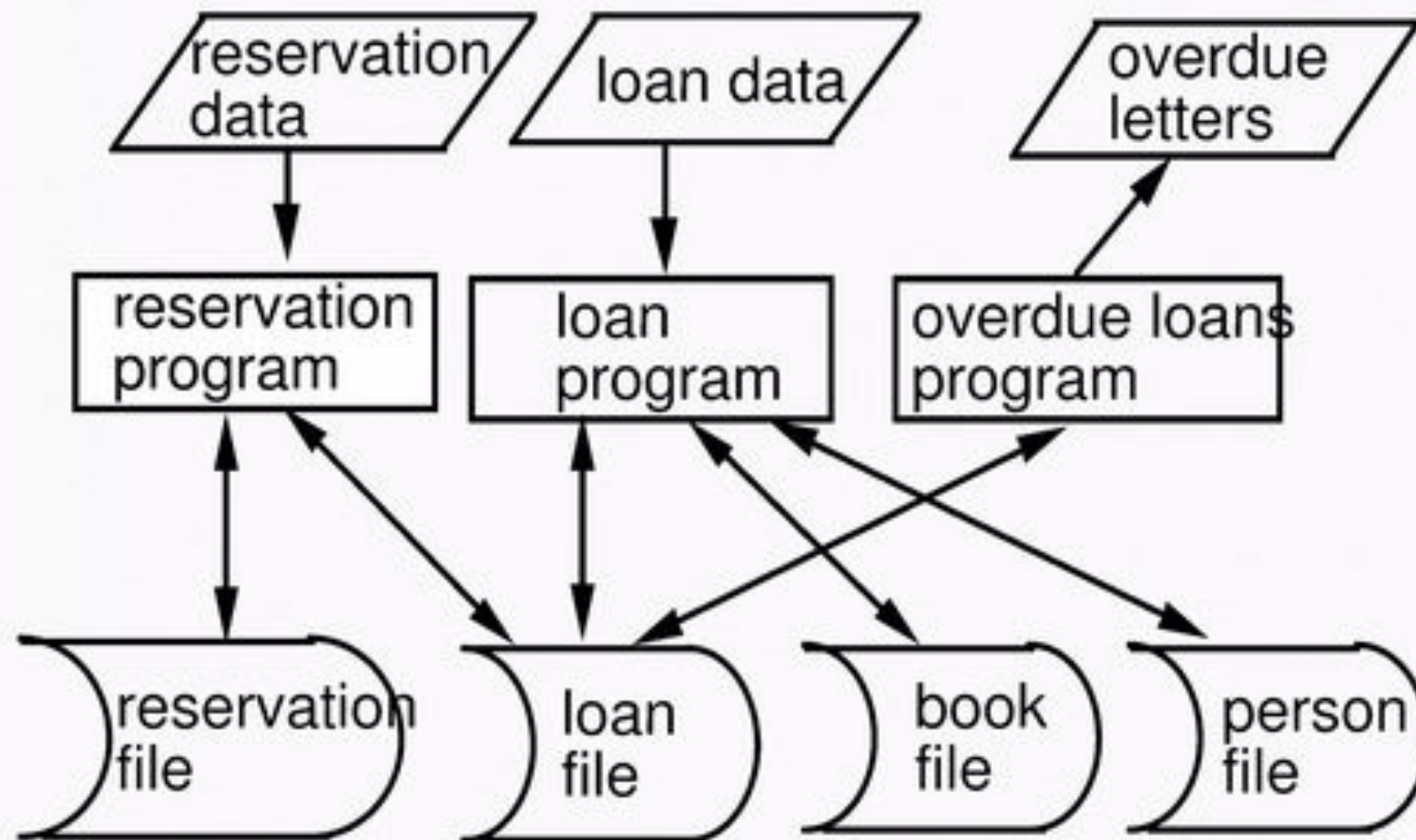
File system architecture

Consider an example of a student's file system. The student file will contain information regarding the student we have a subject file that contains information about the subject and the result file which contains the information regarding the result.

Some fields are duplicated in more than one file, which leads to data redundancy. So to overcome this problem, we need to create a centralized system, i.e. DBMS approach.



Files Dedicated to Application Programs



Disadvantages of file oriented approach

▶ 1) Data redundancy and inconsistency:

- ▶ The same information may be written in several files. This redundancy leads to higher storage and access cost.
- ▶ It may lead data inconsistency that is the various copies of the same data may present at multiple places for example a changed customer address may be reflected in single file but not else where in the system.

▶ 2) Difficulty in accessing data :

- ▶ The conventional file processing system do not allow data to be retrieved in a convenient
- ▶ and efficient manner according to user choice.

▶ **3) Data isolation :**

- ▶ Because data are scattered in various files and files may be in different formats with new application programs to retrieve the appropriate data is difficult.

▶ **4) Integrity Problems:**

- ▶ Developers enforce data validation in the system by adding appropriate code in the various application program.
- ▶ How ever when new constraints are added, it is difficult to change the programs to enforce them.

▶ **5) Atomicity:**

- ▶ It is difficult to ensure atomicity in a file processing system when transaction failure occurs due to power failure, networking problems etc. (atomicity: either all operations of the transaction are reflected properly in the database or non are)

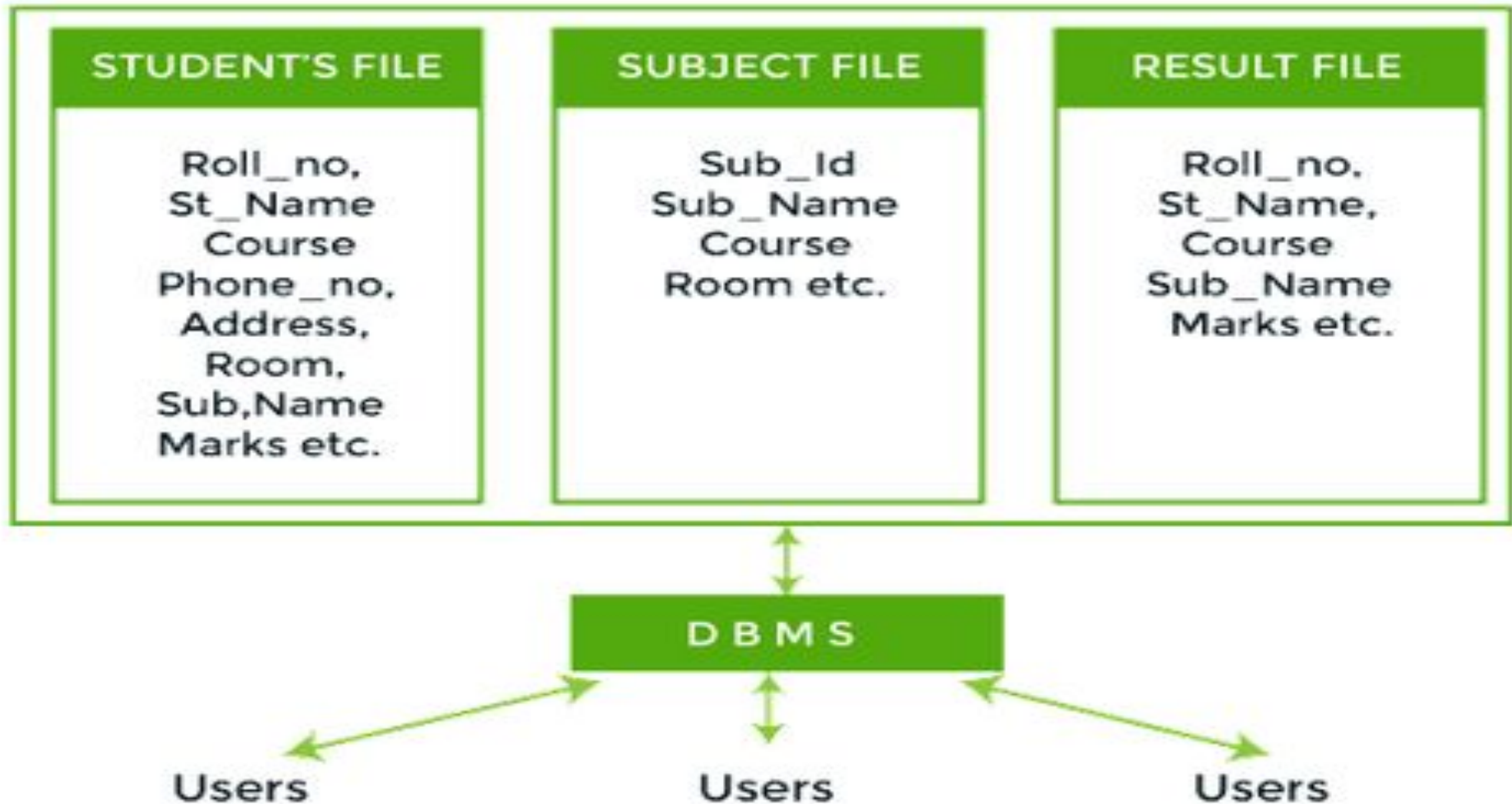
▶ **6) Concurrent access:** In the file processing system it is not possible to access the same file for transaction at same the time.

▶ **7) Security problems:** There is no security provided in file processing system to secure the data from unauthoriz- ed user access.

Why Database:

- ▶ In order to overcome the limitation of a file system, a new approach was required.
- ▶ Hence a database approach emerged. A database is a persistent collection of logically related data. The initial attempts were to provide a centralized collection of data.
- ▶ A database has a self describing nature. It contains not only the data sharing and also integration of data of an organization in a single database.

Data base system



Database defined

- ▶ A database is an organized collection of structured information, or data, typically stored electronically in a computer system.
- ▶ A database is usually controlled by a database management system (DBMS). Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a database system, often shortened to just database.
- ▶ A database is organized collection of related data of an organization stored in formatted way which is shared by multiple users

► **Persistent:**

If data is removed from database due to some explicit request from user to remove.

► **Integrated:**

A database can be a collection of data from different files and when any redundancy among those files are removed from database is said to be integrated data.

► **Sharing Data:**

The data stored in the database can be shared by multiple users simultaneously without affecting the correctness of data.

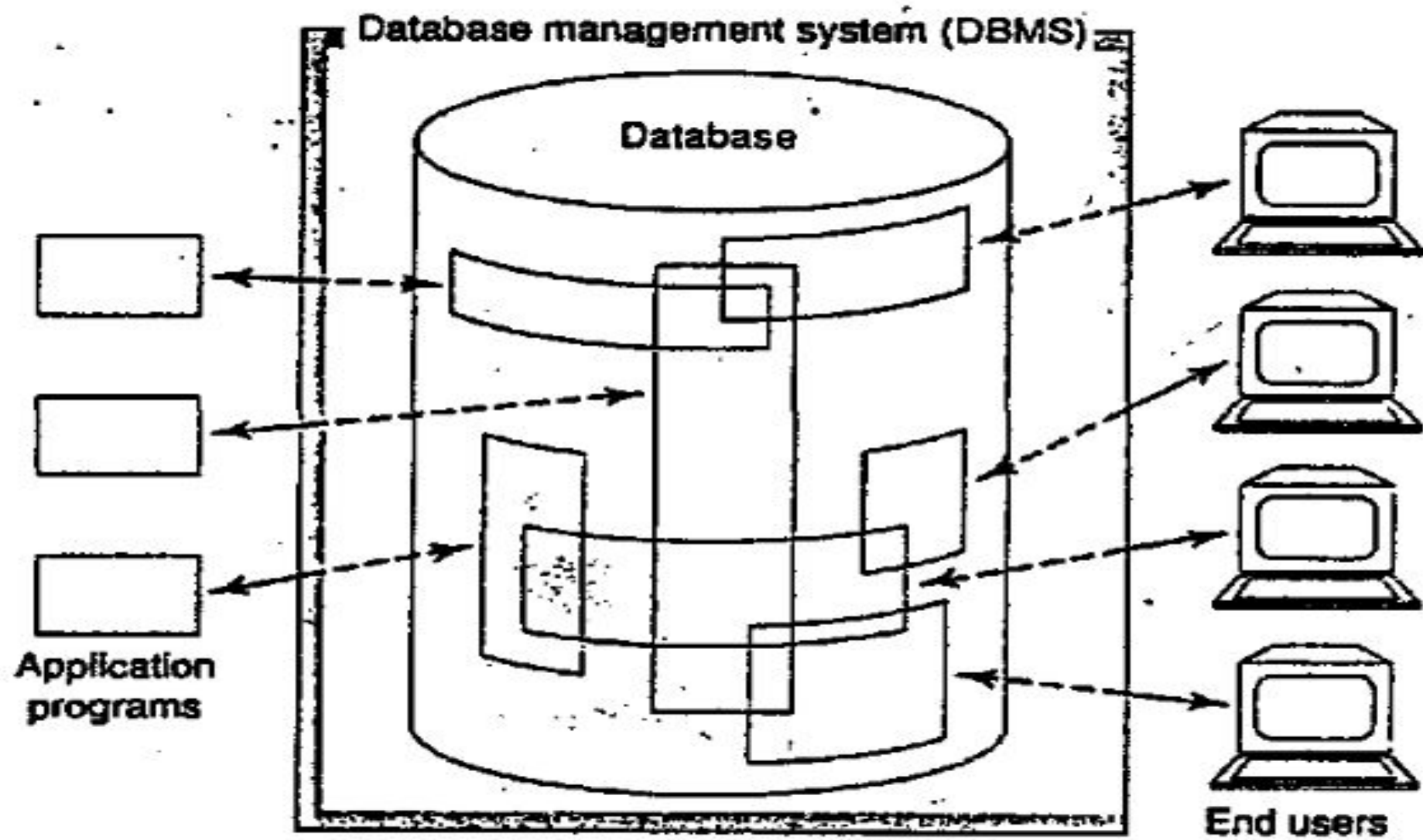


Fig. 1.4 Simplified picture of a database system

Database Applications:

- ▶ Banking: transactions
- ▶ Airlines: reservations, schedules
- ▶ Universities: registration, grades
- ▶ Sales: customers, products, purchases
- ▶ Online retailers: order tracking, customized recommendations
- ▶ Manufacturing: production, inventory, orders, supply chain
- ▶ Human resources: employee records, salaries, tax deductions

The main feature of data in a database are:

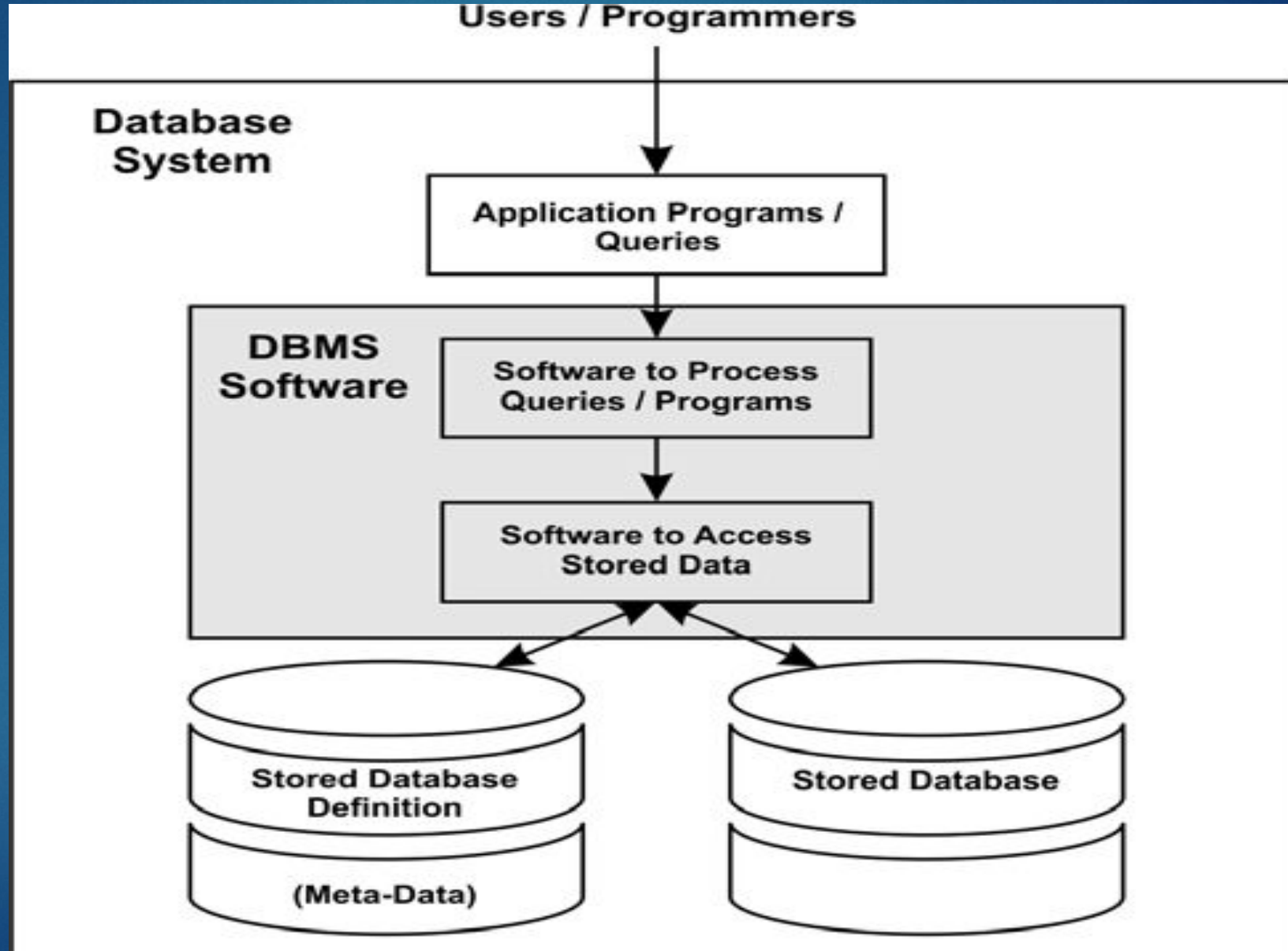


- ▶ 1. It must be well organized
- ▶ 2. It is related
- ▶ 3. It is accessible in a logical order without any difficulty
- ▶ 4. It is stored only once
- ▶ For example consider the roll no, name, address of a student stored in a student file. It is collection
- ▶ of related data with an implicit meaning. Data in the database may be persistent, integrated and shared.

Data base Systems

- ▶ A Database Management System (DBMS) is a software system that is designed to manage and organize data in a structured manner.
- ▶ It allows users to create, modify, and query a database, as well as manage the security and access controls for that database.
- ▶ DBMS provides an environment to store and retrieve the data in convenient and efficient manner.

Database System



Key Features of DBMS

- ▶ **Data organization:** A DBMS allows for the organization and storage of data in a structured manner, making it easy to retrieve and query the data as needed.
- ▶ **Data integrity:** A DBMS provides mechanisms for enforcing data integrity constraints, such as constraints on the values of data and access controls that restrict who can access the data.
- ▶ **Concurrent access:** A DBMS provides mechanisms for controlling concurrent access to the database, to ensure that multiple users can access the data without conflicting with each other.
- ▶ **Data security:** A DBMS provides tools for managing the security of the data, such as controlling access to the data and encrypting sensitive data.
- ▶ **Backup and recovery:** A DBMS provides mechanisms for backing up and recovering the data in the event of a system failure.
- ▶ **Data sharing:** A DBMS allows multiple users to access and share the same data, which can be useful in a collaborative work environment.

differences between DBMS and File systems:

Basis	DBMS Approach	File System Approach
Sharing of data	Due to the centralized approach, data sharing is easy.	Data is distributed in many files, and it may be of different formats, so it isn't easy to share data.
Data Abstraction	DBMS gives an abstract view of data that hides the details.	The file system provides the detail of the data representation and storage of data.
Security and Protection	DBMS provides a good protection mechanism.	It isn't easy to protect a file under the file system.
Recovery Mechanism	DBMS provides a crash recovery mechanism, i.e., DBMS protects the user from system failure.	The file system doesn't have a crash mechanism, i.e., if the system crashes while entering some data, then the content of the file will be lost.
Manipulation Techniques	DBMS contains a wide variety of sophisticated techniques to store and retrieve the data.	The file system can't efficiently store and retrieve the data.
Concurrency Problems	DBMS takes care of Concurrent access of data using some form of locking.	In the File system, concurrent access has many problems like redirecting the file while deleting some information or updating some information.
Data Redundancy and Inconsistency	Due to the centralization of the database, the problems of data redundancy and inconsistency are controlled.	In this, the files and application programs are created by different programmers so that there exists a lot of duplication of data which may lead to inconsistency.

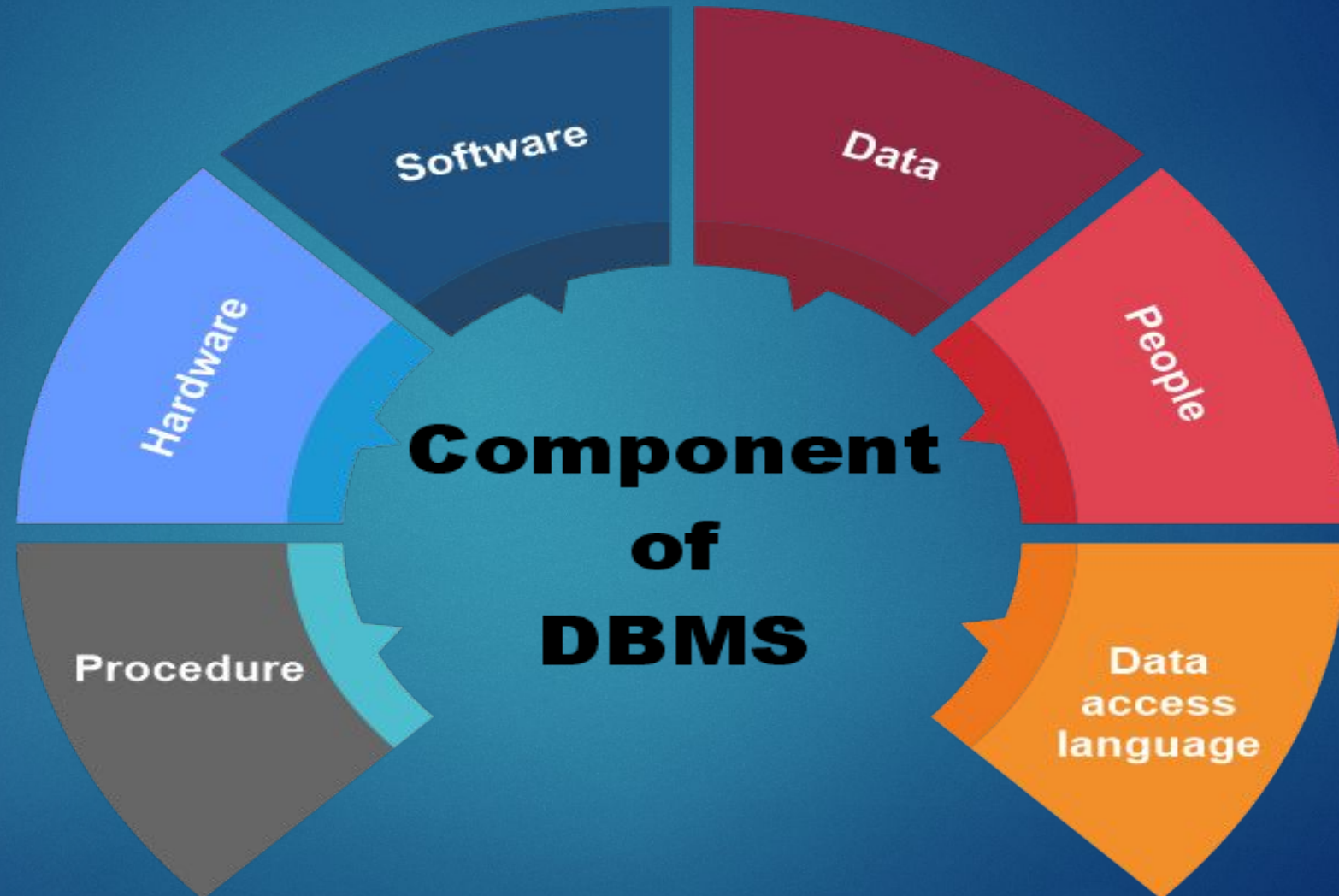
DBMS can be classified into two types:

- ▶ **1. Relational Database Management System (RDBMS)&(OODBMS)**
- ▶ **2. Non-Relational Database Management System (NoSQL or Non-SQL)**
- ▶ **Relational DBMS (RDBMS):** An RDBMS stores data in tables with rows and columns, and uses SQL (Structured Query Language) to manipulate the data.
- ▶ **Object-Oriented DBMS (OODBMS):** An OODBMS stores data as objects, which can be manipulated using object-oriented programming languages.
- ▶ **NoSQL DBMS:** A NoSQL DBMS stores data in non-relational data structures, such as key-value pairs, document-based models, or graph models.

Database Management Systems

- ▶ There are Database Management Systems (DBMS), such as:
 - ▶ • Microsoft SQL Server
 - ▶ • Oracle
 - ▶ • Sybase
 - ▶ • dBase
 - ▶ • Microsoft Access
 - ▶ • MySQL from Sun Microsystems (Oracle)
 - ▶ • DB2 from IBM
 - ▶ SQL Server

Components of DBMS



Components

- ▶ **1. Hardware** : Here the hardware means the physical part of the DBMS. Here the hardware includes output devices like a printer, monitor, etc., and storage devices like a hard disk.
- ▶ **With the help of hardware, the DBMS can access and update the database.**
- ▶ **The server can store a large amount of data, which can be shared with the help of the user's own system.**
- ▶ **The database can be run in any system that ranges from microcomputers to mainframe computers. And this database also provides an interface between the real worlds to the database.**

2. software

- ▶ **Software** :This software has the ability to understand the database accessing language and then convert these languages to real database commands and then execute the database.
- ▶ This is the main component as the total database operation works on a software or application, which provides an easy interface for the user to store, update and delete the data from the database.
- ▶ Some examples of DBMS software include MySQL, Oracle, SQL Server, dBase, FileMaker, Clipper, Foxpro, Microsoft Access, etc.

3. Data

- ▶ The term data means the collection of any raw fact stored in the database. Here the data are any type of raw material from which meaningful information is generated.
- ▶ The main reason behind discovering the database is to create and manage the data within the database.
- ▶ Data is the most important part of the DBMS. Here the database contains the actual data and metadata. Here **metadata** means data about data.
- ▶ For example, when the user stores the data in a database, some data, such as the size of the data, the name of the data, and some data related to the user, are stored within the database. These data are called metadata.

4. Procedures

- ▶ The procedure is a type of general instruction or guidelines for the use of DBMS.
- ▶ The main purpose of the procedure is to guide the user during the management and operation of the database.
- ▶ The procedure of the databases is so similar to the function of the database.

Database Access Language

- ▶ Database Access Language is a simple language that allows users to write commands to perform the desired operations on the data that is stored in the database.
- ▶ Database Access Language is a language used to write commands to access, upsert, and delete data stored in a database.
- ▶ Users can write commands or query the database using Database Access Language before submitting them to the database for execution.
- ▶ Through utilizing the language, users can create new databases and tables, insert data and delete data.
- ▶ **Examples of database languages** are SQL (structured query language), My Access, Oracle, etc. A database language is comprised of two languages.

People

- ▶ The people who control and manage the databases and perform different types of operations on the database in the DBMS.
- ▶ The people include **database administrator, software developer, and End-user.**
- ▶ **Database administrator(DBA)**-database administrator is the one who manages the complete database management system. DBA takes care of the security of the DBMS, its availability, managing the license keys, managing user accounts and access, etc.
- ▶ **Software developer**- This user group is involved in developing and designing the parts of DBMS. They can handle massive quantities of data, modify and edit databases, design and develop new databases, and troubleshoot database issues.
- ▶ **End user** - These days, all modern web or mobile applications store user data. How do you think they do it? Yes, applications are programmed in such a way that they collect user data and store the data on a DBMS system running on their server. End users are the ones who store, retrieve, update and delete data.

The users of the database can be classified into different groups.

- ▶ Native Users
- ▶ Online Users
- ▶ Sophisticated Users
- ▶ Specialized Users
- ▶ Application Users
- ▶ DBA - Database Administrator

Disadvantages of DBMS

- ▶ Increased cost :-Cost of hardware and software, Cost of staff training, Cost of data conversion.
- ▶ 2. Complexity
- ▶ 3. Performance
- ▶ 4. Frequency Upgrade/Replacement Cycles